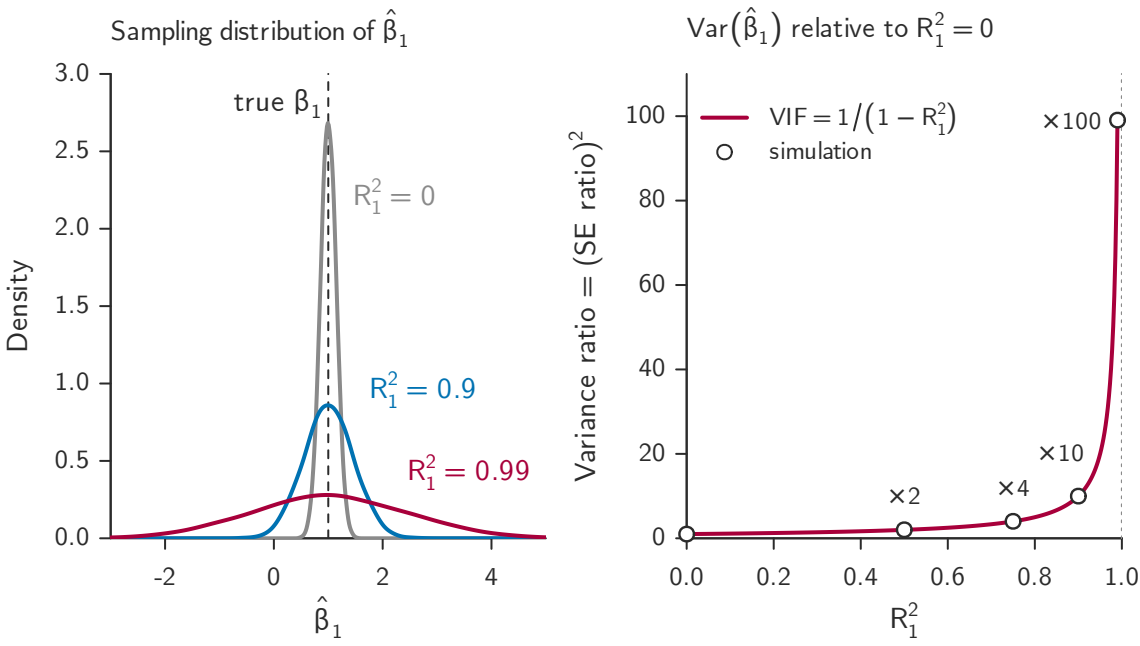
Sampling distribution of $\hat{\beta}_1$
Density
3.0
2.5
2.0
1.5
1.0
0.5
0.0
true β_1
$R_1^2 = 0$
$R_1^2 = 0.9$
$R_1^2 = 0.99$
-2
0
2
4
$\hat{\beta}_1$
$\mathrm{Var}(\hat{\beta}_1)$ relative to $R_1^2 = 0$
Variance ratio = (SE ratio)^2
100
80
60
40
20
0
$\mathrm{VIF} = 1/(1 - R_1^2)$
simulation
×100
×10
×4
×2
0.0
0.2
0.4
0.6
0.8
1.0
R_1^2